```
        padded[cx:-cx, cy:-cy] = im1

    # Convolve entire image
    for r in range(padded.shape[0] - (filt.shape[0]-1)):
        for c in range(padded.shape[1] - (filt.shape[1]-1)):
            convolved[r,c] = np.sum(np.multiply(filt,
                padded[r:r+filt.shape[0], c:c+filt.shape[1]]))

    return convolved

# Four-loop convolution implementation
def convolve4l(im1, filt):
    # Preprocessing (padding) implementation not shown

    # Convolve entire image with four nested loops
    for r in range(padded.shape[0] - (filt.shape[0]-1)):
        for c in range(padded.shape[1] - (filt.shape[1]-1)):
            filtered = 0
            for i in range(filt.shape[0]):
                for j in range(filt.shape[1]):
                    filtered += filt[i,j] * padded[r + i, j + c]
            convolved[r,c] = filtered
    return convolved
```

## Results

Here is my selfie and a convolution with my implementation

Grayscale Selfie

Convolved Selfie

## And here is a comparison to the scipy version

Scipy Convolution Result

## Commentary